Weather Analytics API Documentation

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1. API Overview

- **Project**: Weather Analytics Web API (COMP3011 CW1)
- **Version**: 0.1.0
- **Framework**: FastAPI
- **Response** Format: JSON

This API manages city locations, stores weather observations retrieved from OpenWeatherMap, and provides analytical endpoints such as temperature and humidity statistics and trends.

Key Features

- Location management with CRUD operations
- Weather observation storage
- Featured locations dashboard
- Temperature and humidity statistics
- Temperature trend analytics

Architecture

```
Next.js Frontend
      ↓
FastAPI REST API
      ↓
PostgreSQL Database
      ↓
OpenWeatherMap API
```

2. Base URL

- **Local Development**: http://localhost:8000
- **Swagger UI**: http://localhost:8000/docs
- **OpenAPI Schema**: http://localhost:8000/openapi.json

3. Endpoint Summary

Method	Endpoint	Description
GET	/health	API health check
GET	/home	Dashboard data
GET	/locations	List locations
POST	/locations	Create location
GET	/locations/{id}	Get location
PATCH	/locations/{id}	Update location
DELETE	/locations/{id}	Delete location
GET	/locations/search	Search locations
POST	/observations/fetch	Fetch weather from OpenWeatherMap
GET	/observations/latest	Get latest observation
GET	/observations	List observations
GET	/analytics/temperature-stats	Temperature statistics
GET	/analytics/humidity-stats	Humidity statistics
GET	/analytics/temperature-trend	Temperature trend

4. Health Endpoint

GET /health

Check whether the API server is running.

Response Example

```
{
  "status": "ok"
}
```

5. Home Dashboard Endpoint

GET /home

Returns dashboard data including featured locations and latest weather observations.

Query Parameters

Name	Type	Required	Description
refresh	boolean	optional	refresh observation data

Example Response

```
{
  "generated_at": "2026-03-09T10:30:00",
  "featured": [],
  "list": []
}
```

6. Locations API

GET /locations

Returns a list of all stored locations.

Query Parameters

Name	Type	Required	Description
featured	boolean	optional	filter featured locations

Response

```
200 OK
```

Example

```
[
 {
  "id":1,
  "name":"London",
  "country_code":"GB",
  "lat":51.5074,
  "lon":-0.1278,
  "is_featured":true,
  "display_order":1,
  "is_active":true
 }
]
```

POST /locations

Creates a new location.

Request Body

```
{
 "name": "London",
 "country_code": "GB",
 "lat": 51.5074,
 "lon": -0.1278
}
```

Response

```
201 Created
```

GET /locations/{location_id}

Returns a single location.

Response

```
200 OK
```

PATCH /locations/{location_id}

Updates location fields such as featured status or display order.

Example request

```
{
 "is_featured": true,
 "display_order": 1
}
```

DELETE /locations/{location_id}

Soft delete (deactivates) a location.

Response

```
204 No Content
```

7. Location Search

GET /locations/search

Search locations using a keyword.

Query Parameters

Name	Type	Required	Description
q	string	yes	search query
limit	integer	optional	max results

Example

```
GET /locations/search?q=London
```

8. Observations API

POST /observations/fetch

Fetches current weather data from OpenWeatherMap and stores it.

Query Parameters

Name	Type	Required	Description
location_id	integer	yes	location identifier

Example

```
POST /observations/fetch?location_id=1
```

GET /observations/latest

Returns the most recent observation for a location.

Example

```
GET /observations/latest?location_id=1
```

GET /observations

Lists stored observations with optional filters:

Query Parameters

Name	Type	Required	Description
location_id	integer	yes	location identifier
from_dt	datetime	optional	start time
to_dt	datetime	optional	end time
limit	integer	optional	max result
offset	integer	optional	pagination

9. Analytics API

GET /analytics/temperature-stats

Returns average, minimum, and maximum temperature.

Query Parameters

Name	Type	Required	Description
location_id	integer	yes	location identifier
days	integer	optional	duration (days)

Response example

```
{
 "location_id":1,
 "avg_temp":18.2,
 "min_temp":14.1,
 "max_temp":22.5,
 "count":24
}
```

GET /analytics/humidity-stats

Returns humidity statistics.

Response example

```
{
 "location_id":1,
 "avg_humidity":72.1,
 "min_humidity":60,
 "max_humidity":88,
 "count":24
}
```

GET /analytics/temperature-trend

Returns daily average temperature trend over a specified number of days.

Query Parameters

Name	Type	Required	Description
location_id	integer	yes	location identifier
days	integer	optional	duration (days)

Response example

```
{
 "location_id":1,
 "days":7,
 "data":[
   {"date":"2026-03-01","avg_temp":10.3}
 ]
}
```

10. Error Handling

Standard HTTP status codes are used:

Code	Meaning
200	Successful request
201	Resource created
204	Resource deleted
422	Validation error

Validation errors follow FastAPI standard schema.

11. Data Source

Weather data is retrieved from the OpenWeatherMap API.

https://openweathermap.org/api

12. Notes

All API responses are returned in **JSON format**

The APIs is implemented using the **FastAPI framework**